

Software Testing Mastery

A 4-Month Industry-Aligned Online Course

From fundamentals to job-ready skills — a practical, structured learning journey designed for early professionals, graduate students, and career transitioners entering the world of software quality assurance.

4 MONTHS

15–16 WEEKS

CAPSTONE INCLUDED

LIVE MENTORING

Why This Course Matters

Software Testing is the Backbone of Quality Engineering

The Industry Reality

Over 80% of software failures are attributed to inadequate testing. As systems grow more complex, organizations need skilled QA engineers who can prevent costly defects before they reach production.

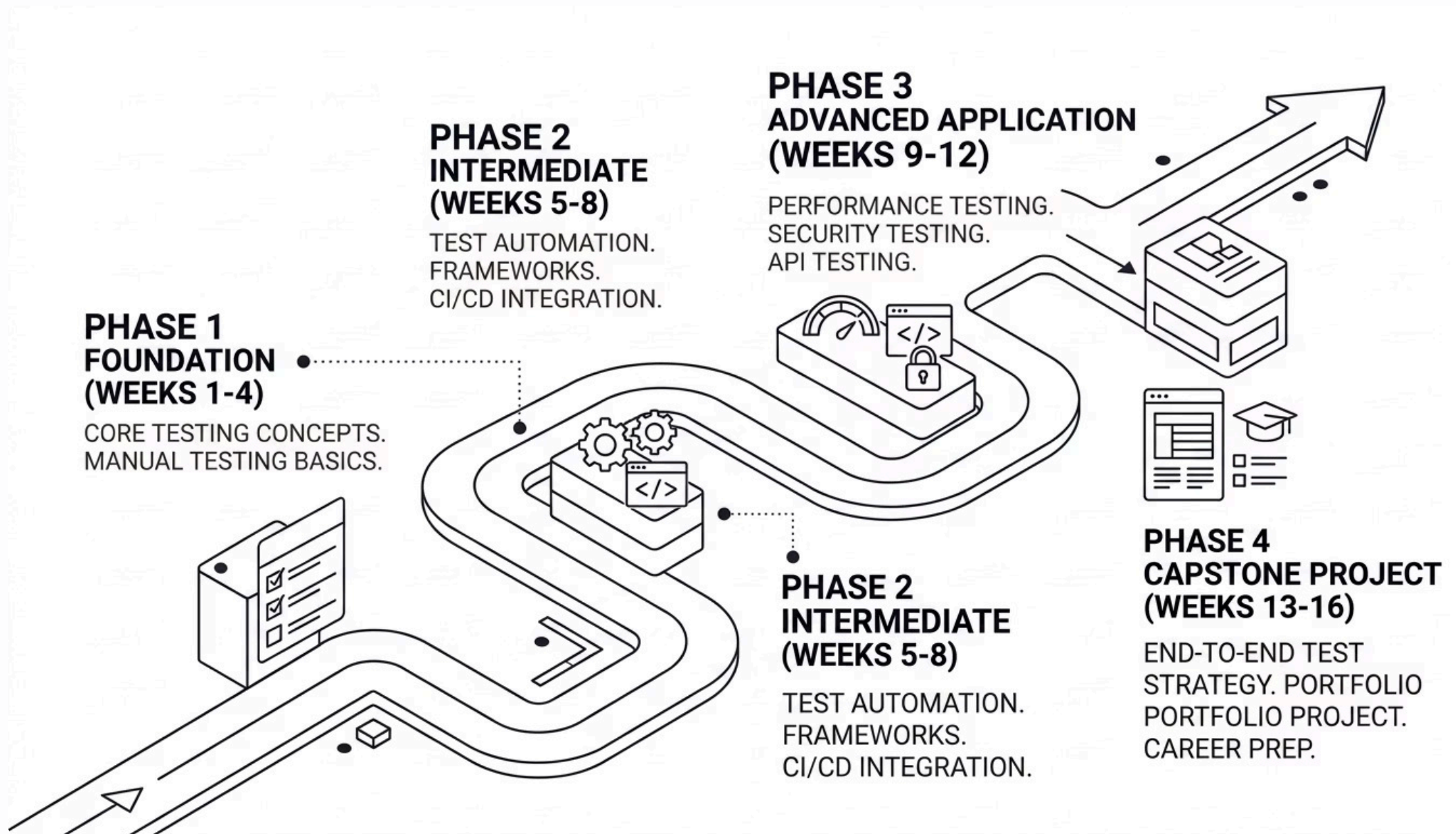
- Global QA market exceeds \$40B annually
- Testing roles among top 10 fastest-growing tech jobs
- Automation demand up 60% in 3 years

What This Course Delivers

This program bridges the gap between classroom theory and workplace expectations. You'll graduate with hands-on tool experience, real project exposure, and a portfolio that speaks to hiring managers.

- Industry-standard tools and workflows
- Real-world case studies and scenarios
- Mentored learning with direct feedback
- A capstone project to anchor your portfolio

Your 4-Month Learning Roadmap



Each phase builds deliberately on the last — starting with a solid conceptual foundation, advancing through automation and specialized testing, and culminating in a capstone project that mirrors real industry deliverables.

Program Format

How the Course Is Structured Each Week



Weekend Recorded Session

A concept-focused recorded session released each weekend. Watch at your own pace, pause and rewatch as needed, and arrive at mentoring sessions with questions ready.



Live 1:1 or Small-Group Mentoring

A weekly live session for doubt-clearing, case discussions, and real-time feedback on your assignments. Small groups ensure personalized attention and peer learning.



Weekly Assignments & Mini Projects

Each week ends with a practical exercise or mini project that reinforces core concepts and builds your growing portfolio of work samples.



Capstone Project (Weeks 13–16)

A culminating end-to-end test project on a real-world application, simulating a full QA engagement — from test planning to final report and demo presentation.

Foundation Phase — Weeks 1-4

Build the conceptual bedrock of software testing. Understand the **why** before the **how** — covering testing principles, the SDLC, test design, and your first hands-on tool exposure.

1

Week 1

Introduction to Software Testing

2

Week 2

Test Design Techniques

3

Week 3

Test Planning & Management

4

Week 4

Defect Lifecycle & Bug Reporting



Learning Mix: 80% core concepts, 20% hands-on practice with Jira, TestRail, and basic test case writing exercises.

Foundation Phase — Week 1

Introduction to Software Testing

Learning Objectives

- Define software testing and explain its purpose in the SDLC
- Distinguish between verification and validation
- Compare and contrast manual vs. automated testing
- Understand the seven principles of software testing
- Identify different testing levels: unit, integration, system, acceptance
- Recognize the cost of defects at different SDLC stages

Key Concepts Covered

- **SDLC & STLC** — how testing fits into the development lifecycle
- **Testing Types** — functional, non-functional, regression, smoke
- **Testing Levels** — unit → integration → system → UAT
- **QA vs QC vs Testing** — critical distinctions often confused
- **Shift-Left Testing** — testing earlier to reduce cost of failure
- **Industry Context** — how modern teams structure QA roles

Practical Exercise

Analyze a real defect report from an open-source project and classify the bug by type, severity, and priority.

Foundation Phase — Week 2

Test Design Techniques

Learning Objectives

- Apply black-box and white-box design techniques
- Write effective test cases using equivalence partitioning
- Use boundary value analysis to catch edge-case defects
- Create decision tables for complex logic scenarios
- Design state transition diagrams for workflow-based systems
- Measure test coverage using basic metrics

Tools Introduced

TESTRAIL

EXCEL / GOOGLE SHEETS

MIND MAPS

Key Concepts Covered

- **Equivalence Partitioning** — divide inputs into valid/invalid classes
- **Boundary Value Analysis** — test at the edges of input ranges
- **Decision Tables** — map conditions to expected outcomes
- **State Transition Testing** — model system behavior across states
- **Use Case Testing** — derive tests from user scenarios
- **Exploratory Testing** — structured free-form investigation

Mini Assignment

Write 15 test cases for a login form using at least three design techniques. Document your coverage rationale.

Foundation Phase — Weeks 3 & 4

Test Planning, Management & Defect Lifecycle

Week 3 — Test Planning

- Writing a Test Plan document
- Scope, objectives, and risk identification
- Test estimation techniques (point-based, expert judgment)
- Entry and exit criteria
- Test environment setup basics
- Roles and responsibilities in a QA team

Week 4 — Defect Lifecycle

- Anatomy of a great bug report
- Defect lifecycle states: New → Assigned → Fixed → Verified → Closed
- Severity vs. Priority — and why the difference matters
- Bug tracking with Jira — hands-on workflow
- Defect metrics and reporting to stakeholders
- Root cause analysis fundamentals

- ✔ Phase 1 Outcome: You can write structured test plans, design effective test cases, and file professional-quality bug reports using industry-standard tools.

PHASE 2

Intermediate Phase — Weeks 5–8

Level up from manual testing to the automated workflows that dominate modern QA teams. Learn Selenium, build reusable frameworks, and plug your tests into CI/CD pipelines.



Week 5

Intro to Test Automation & Selenium



Week 6

Automation Frameworks



Week 7

API Testing with Postman



Week 8

CI/CD Integration & DevOps QA

Introduction to Test Automation & Selenium

Learning Objectives

- Understand when automation adds value vs. when it doesn't
- Set up a Selenium WebDriver environment from scratch
- Write basic test scripts using Java or Python
- Interact with web elements: buttons, forms, dropdowns
- Apply XPath and CSS selectors effectively
- Run and interpret your first automated test suite

Key Concepts & Tools

- **Automation ROI** — calculating when to automate
- **Selenium WebDriver** — architecture and core components
- **Locator Strategies** — ID, Name, XPath, CSS Selector
- **Waits** — implicit, explicit, and fluent waits explained
- **TestNG / JUnit** — structuring and running test suites
- **WebDriverManager** — driver setup without manual downloads

SELENIUM

JAVA / PYTHON

TESTNG

INTELLIJ / VS CODE

Mini Project

Automate an end-to-end login and checkout flow on a demo e-commerce site (e.g., SauceDemo or DemoBlaze).

Intermediate Phase — Weeks 6, 7 & 8

Frameworks, API Testing & CI/CD Integration

Week 6 — Automation Frameworks

- **Page Object Model (POM)** — separating test logic from UI structure
- Data-Driven Testing with Excel/CSV inputs
- Keyword-Driven and Hybrid Framework design
- Reporting with ExtentReports or Allure
- Handling dynamic elements and iframes
- Screenshot capture on test failures

Week 7 — API Testing with Postman & RestAssured

- REST API fundamentals — endpoints, methods, status codes
- Writing API tests in Postman: GET, POST, PUT, DELETE
- JSON schema validation and response assertions
- Automated API testing with RestAssured (Java)
- Environment variables and collections in Postman
- Contract testing concepts

Week 8 — CI/CD Integration & DevOps QA

- Jenkins setup and pipeline basics for QA
- Running Selenium tests in a CI pipeline
- GitHub Actions for automated test triggers
- Test reporting integration with CI dashboards
- Dockerized test environments introduction
- QA's evolving role in Agile and DevOps teams

PHASE 3

Advanced & Application Phase — Weeks 9–12

Expand your specialization into performance, security, and mobile testing — the high-demand skills that separate mid-level testers from senior QA engineers.

Week 9

Performance & Load Testing

JMeter, load vs. stress vs. soak testing, performance benchmarks, and bottleneck identification

Week 10

Security Testing Fundamentals

OWASP Top 10, SQL injection, XSS, BURP Suite basics, security test planning

Week 11

Mobile & Cross-Browser Testing

Appium for mobile automation, BrowserStack, responsive testing, device lab strategy

Week 12

Test Strategy & Agile QA

Risk-based testing, Agile ceremonies for QA, BDD with Cucumber, test strategy documentation

Performance & Load Testing with JMeter

Learning Objectives

- Distinguish between load, stress, spike, and soak testing
- Design realistic load test scenarios from user stories
- Configure JMeter test plans with thread groups
- Define performance SLAs and acceptance thresholds
- Identify application bottlenecks using JMeter reports
- Communicate performance findings to development teams

Key Concepts & Tools

- **Apache JMeter** — test plan structure, listeners, and samplers
- **Throughput & Latency** — key metrics and how to interpret them
- **Think Time & Ramp-Up** — simulating realistic user behavior
- **Assertions** — validating response content under load
- **BlazeMeter / Gatling** — cloud-based and code-first alternatives
- **Reporting** — HTML dashboards and summary statistics

APACHE JMETER

BLAZEMETER

GATLING

Mini Project

Run a load test on a public demo API simulating 100 concurrent users. Document findings and identify the breaking point.

Security, Mobile & Agile QA Strategy

1

Week 10 — Security Testing

- OWASP Top 10 vulnerabilities explained for testers
- SQL Injection and XSS testing techniques
- Authentication and authorization testing
- BURP Suite Community Edition — intercepting requests
- OWASP ZAP for automated vulnerability scanning
- Writing a security test report for developers

2

Week 11 — Mobile & Cross-Browser

- Mobile testing challenges vs. web testing
- Appium architecture and setup for Android/iOS
- BrowserStack and Sauce Labs for device coverage
- Responsive design testing methodologies
- Accessibility testing basics (WCAG 2.1)
- Real device vs. emulator trade-offs

3

Week 12 — Agile QA & BDD

- QA's role in Scrum: sprint planning, retrospectives
- Behavior-Driven Development with Cucumber
- Writing Gherkin scenarios: Given/When/Then
- Risk-based test prioritization frameworks
- Test strategy vs. test plan — knowing the difference
- Metrics that matter to QA managers and stakeholders

Capstone Project — Weeks 13–16

Project Brief

Apply everything you've learned in a realistic, end-to-end QA engagement on a multi-feature web application. Your capstone mirrors what junior QA engineers deliver in their first 90 days on the job.

Deliverables

- Test Plan document (scope, risks, strategy)
- Test cases — manual and automated (Selenium)
- API test collection in Postman
- Performance test report (JMeter)
- Bug reports filed in Jira
- Final test summary report
- 15-minute recorded demo presentation

Weekly Capstone Milestones

01

Week 13

Requirements analysis, test plan, and risk assessment

02

Week 14

Manual test execution, bug reporting, and defect triage

03

Week 15

Automation suite development and API/performance testing

04

Week 16

Final report, demo preparation, and mentor review

Real-World Case Study #1: E-Commerce Platform Launch

The Business Problem

A mid-sized U.S. retail company was launching a revamped e-commerce platform ahead of Black Friday. Previous year's launch had resulted in checkout failures affecting 12% of users, costing an estimated \$800K in lost revenue in a single day.

Role of Early Professionals

Junior QA engineers owned the regression test suite execution, wrote automated smoke tests for the checkout flow, and maintained the Jira bug board during the final two-week sprint. Their work enabled nightly regression cycles that senior engineers could review each morning.

How Course Concepts Were Applied

- **Test Design** — equivalence partitioning for payment method combinations
- **Selenium Automation** — 200+ automated regression tests on the cart and checkout
- **API Testing** — validating payment gateway and inventory API responses
- **Performance Testing** — JMeter load tests simulating 5,000 concurrent users
- **CI/CD** — tests triggered on every pull request via Jenkins pipeline

Business Impact

Checkout failure rate dropped from 12% to **under 0.3%**. The automated suite caught 47 critical defects pre-launch. The company reported a **23% increase** in Black Friday revenue vs. the prior year.

Real-World Case Study #2: Banking App Security Audit

The Business Problem

A regional digital bank was preparing for a compliance audit under PCI-DSS standards. Their mobile banking app had not undergone systematic security testing, and the audit was scheduled in 8 weeks. Any failure could result in regulatory fines and reputational damage.

Tools & Technologies Used

OWASP ZAP

BURP SUITE

APPIUM

POSTMAN

JIRA

How Course Concepts Were Applied

- **Security Testing** — OWASP Top 10 checklist applied systematically
- **API Testing** — validating token expiry, authorization headers, and data exposure
- **Mobile Testing** — Appium scripts for session management and data leakage
- **Defect Management** — all security findings triaged and tracked in Jira
- **Test Reporting** — executive-ready security test summary for compliance officers

Business Impact & Role of Early Professionals

Junior testers identified **14 medium-risk and 3 high-risk vulnerabilities** before the audit date. All were resolved, and the bank passed the audit with zero critical findings. Early professionals gained direct exposure to regulated-industry QA practices.

Real-World Case Study #3: SaaS Platform CI/CD Quality Gate

The Business Problem

A B2B SaaS startup was releasing updates to their project management tool twice per week. Rapid deployment cycles meant that manual testing couldn't keep pace — a critical bug slipped into production three times in one quarter, causing customer churn and a surge in support tickets.

Role of Early Professionals

Junior QA engineers built and maintained the automated regression suite, wrote Gherkin feature files for BDD acceptance tests, and monitored test dashboards after each deployment. They held the "quality gate" — nothing merged to main without a green pipeline.

How Course Concepts Were Applied

- **Automation Frameworks** — Page Object Model for maintainable Selenium tests
- **BDD / Cucumber** — acceptance criteria written as Gherkin scenarios
- **CI/CD Integration** — GitHub Actions pipeline with mandatory test gates
- **API Testing** — RestAssured contract tests for every backend endpoint
- **Test Reporting** — Allure reports auto-published to team Slack channel

Business Impact

Production defect rate dropped by **78% in two months**. Deployment confidence increased — release cycles accelerated from bi-weekly to **daily**. Customer churn attributed to bugs fell to near zero in the following quarter.

The Full QA Tool Belt You'll Master

Test Management

- Jira – bug tracking & project management
- TestRail – test case management
- Zephyr Scale – test cycles in Jira

UI Automation

- Selenium WebDriver
- Cypress (intro)
- Playwright (intro)
- TestNG / JUnit

API Testing

- Postman – manual & automated
- RestAssured (Java)
- Newman (CLI runner)

Performance

- Apache JMeter
- BlazeMeter (cloud)
- Gatling (intro)

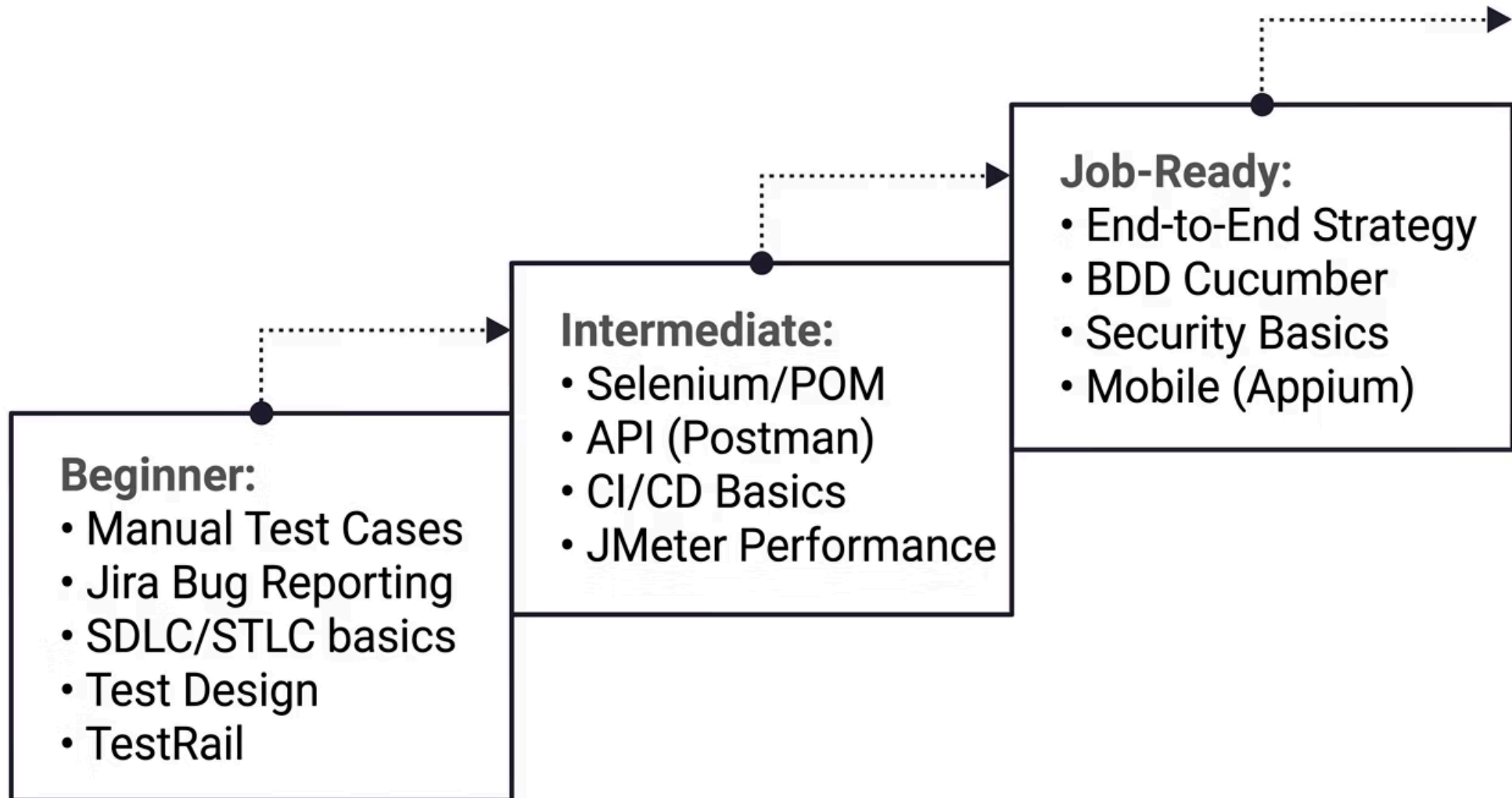
Security

- OWASP ZAP
- BURP Suite Community

DevOps & CI/CD

- Jenkins
- GitHub Actions
- Docker (basics)
- Allure / ExtentReports

Your Progression: Beginner → Intermediate → Job-Ready



Each phase of this course directly corresponds to a maturity level. By Week 12, you'll have the skills to take on an entry-level SDET or QA Engineer role. The capstone project closes any remaining gap between "learned" and "demonstrated."

Career Alignment

Job Roles This Course Prepares You For



QA Engineer

Design and execute manual and automated test suites for web and mobile applications. The most direct entry-point role for course graduates. Median US salary: **\$75K–\$95K**.



QA Analyst / Test Analyst

Focus on requirements analysis, test planning, and defect management. Ideal for learners with strong analytical backgrounds transitioning into QA. Median US salary: **\$70K–\$90K**.



SDET (Software Dev Engineer in Test)

Write production-quality automation code alongside developers. Requires strong programming skills — this course provides the automation foundation. Median US salary: **\$95K–\$120K**.



Performance Test Engineer

Specialize in load and stress testing to ensure applications meet scalability SLAs. High demand in fintech, e-commerce, and enterprise SaaS. Median US salary: **\$90K–\$115K**.

Key Takeaways

What You'll Walk Away With

1 A Structured, Industry-Aligned Skill Set

From test design principles to automation frameworks and CI/CD integration — every skill taught maps directly to what employers expect from junior QA engineers in 2024 and beyond.

3 Confidence With Industry-Standard Tools

Hands-on experience with Jira, Selenium, Postman, JMeter, BURP Suite, Jenkins, and more. Tools that appear in nearly every QA job description in the market today.

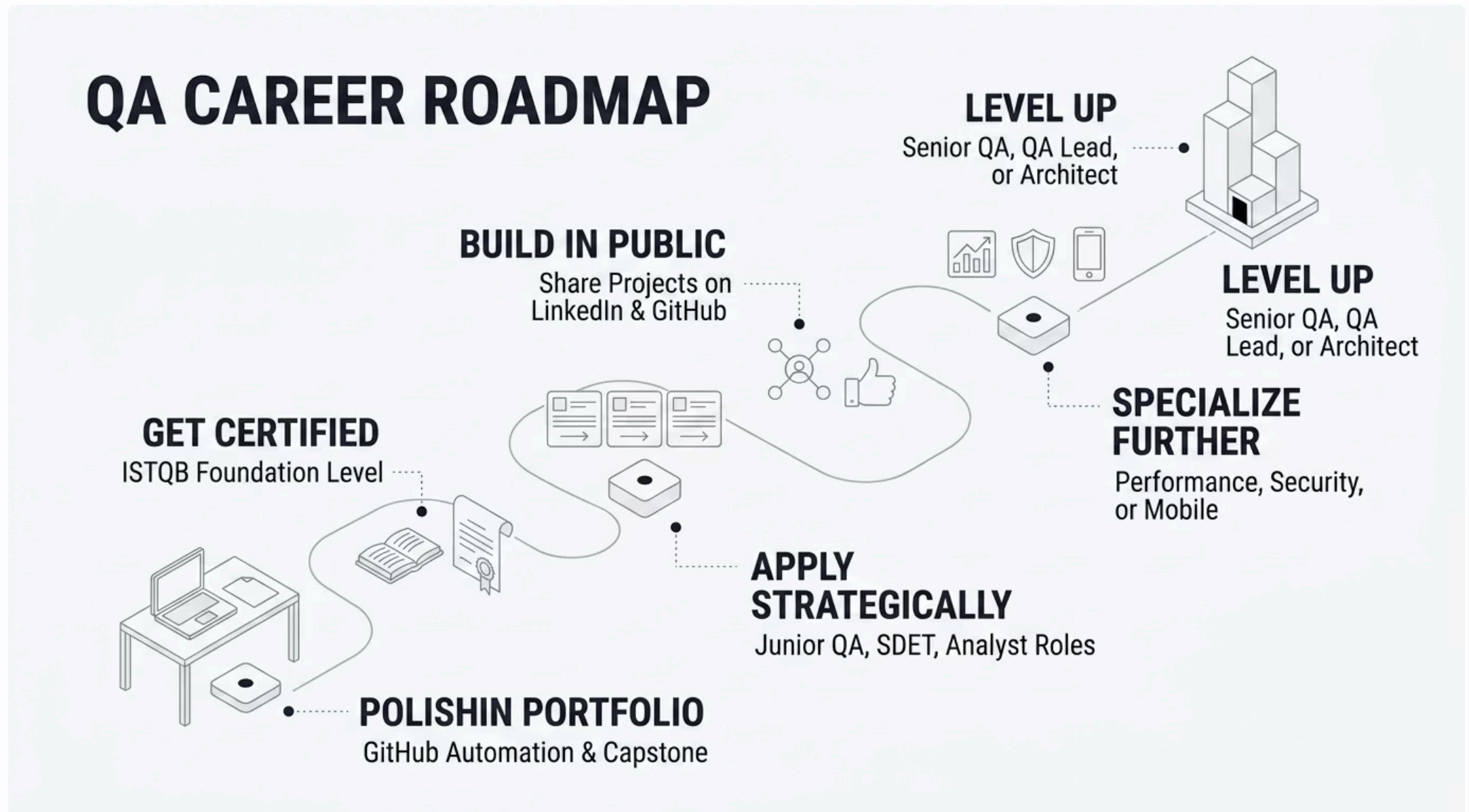
2 A Portfolio of Real Deliverables

You'll have test plans, automation scripts, API collections, JMeter reports, and a complete capstone project to show interviewers — not just a certificate, but evidence of capability.

4 Mentored Growth With Personalized Feedback

Weekly live sessions mean you're never stuck for long. Instructors provide direct feedback on your work, helping you grow faster than self-paced learning alone.

Your Next Steps After Graduation



- ✓ Recommended first certification after this course: **ISTQB Certified Tester Foundation Level (CTFL)** – widely recognized, employer-valued, and directly aligned with the concepts you've mastered.